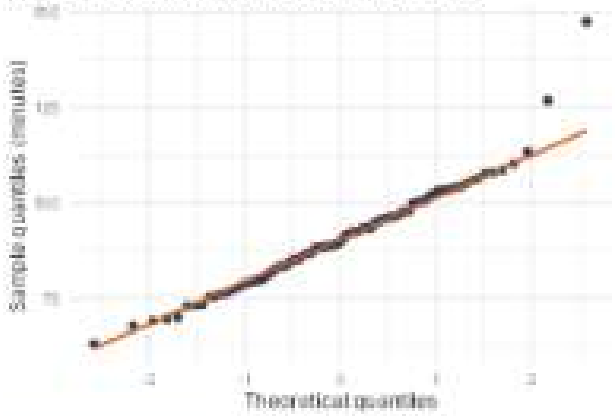
DSA06 - Data Handling and Visualization | Activity 5: Normality Detection using Q-Q Plots | Slot C

R.No: 32 Reg. No: 192424278 Name: SHEIK MOHAMMED AFNAN A

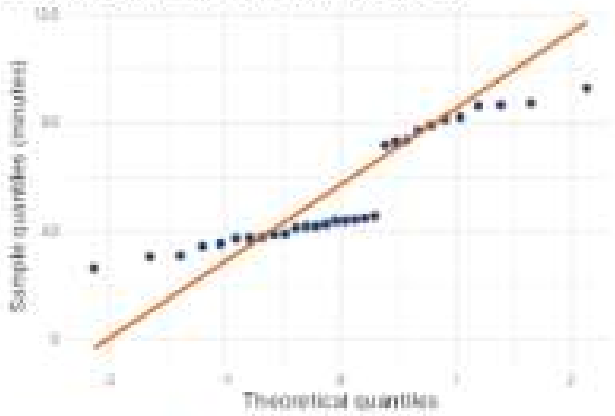
Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Chart 1: lab experiment duration (minutes), n = 100



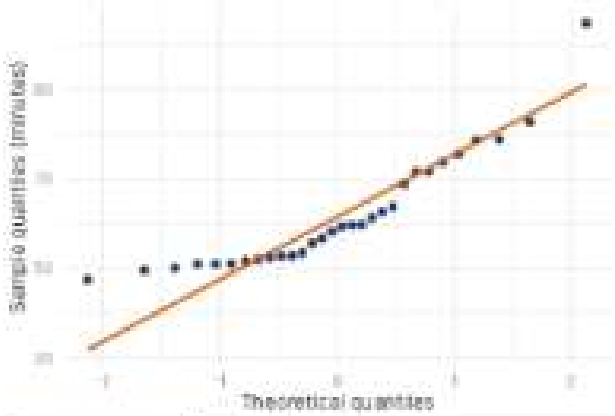
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 2: lab experiment duration (minutes), n = 30



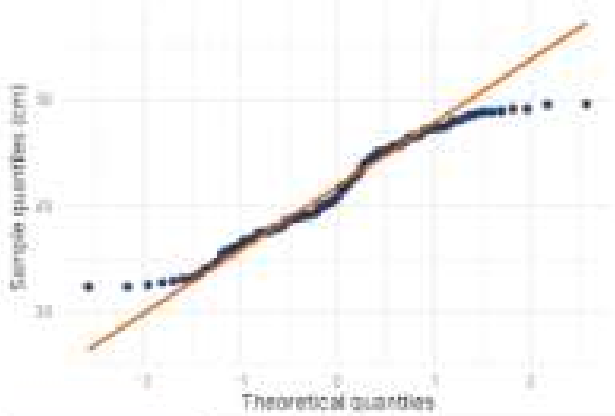
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: lab experiment duration (minutes), n = 30



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 4: plant height (cm), n = 100



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Part B (4 marks)

B1. Chart 1 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

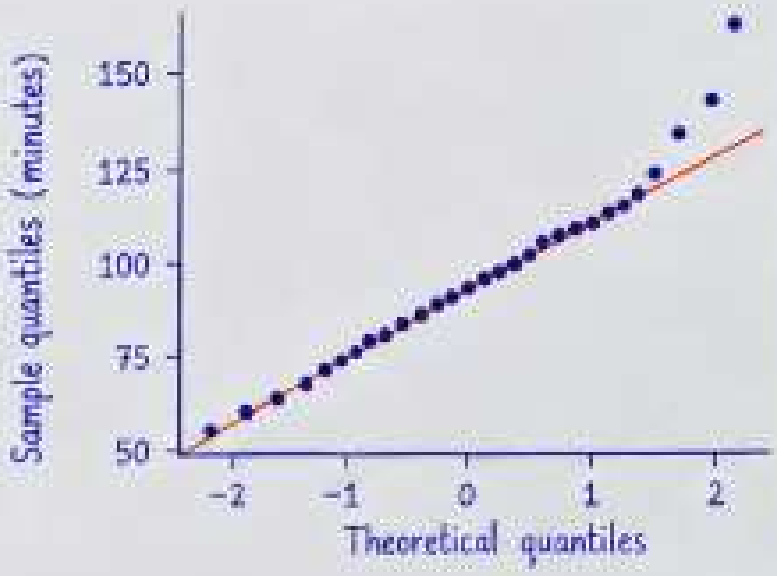
DSA0601 - Data Handling and Visualization

Activity 5 : Normality Detection using Q-Q Plots | Slot C

Reg.No. : 192424278	Name : SHEIK MOHAMMED AFNAN A	R.No : 32

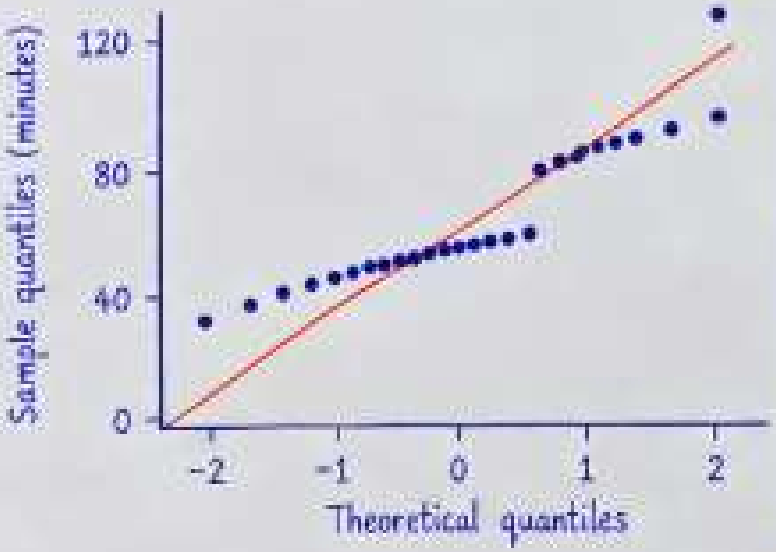
Each chart is a normal Q-Q plot made in R (qqnorm and qqline). Look at the pattern of points and decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Non-normal with outliers. Then write the evidence you see.

Chart 1 : lab experiment duration (minutes), n = 100



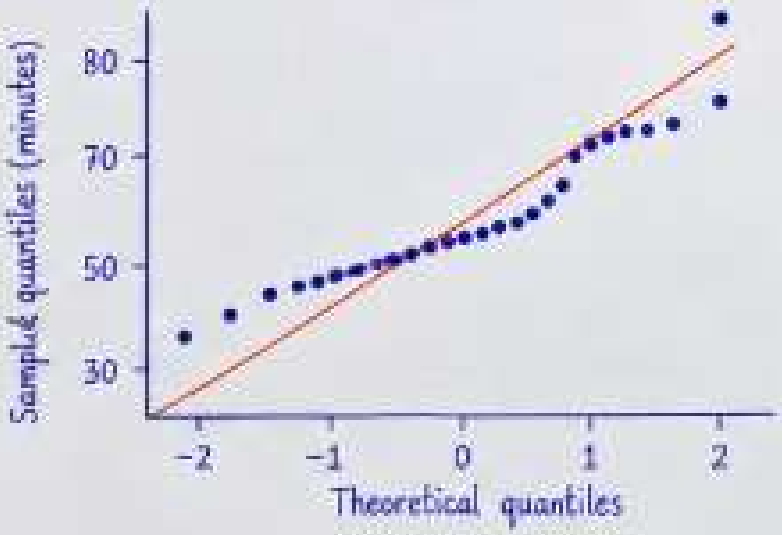
Normal? (Yes / No)	No
Shape (from the list)	Right-skewed
Evidence in the plot	Most points follow the line but the upper tail is above the line (high outliers).

Chart 2 : lab experiment duration (minutes), n = 30



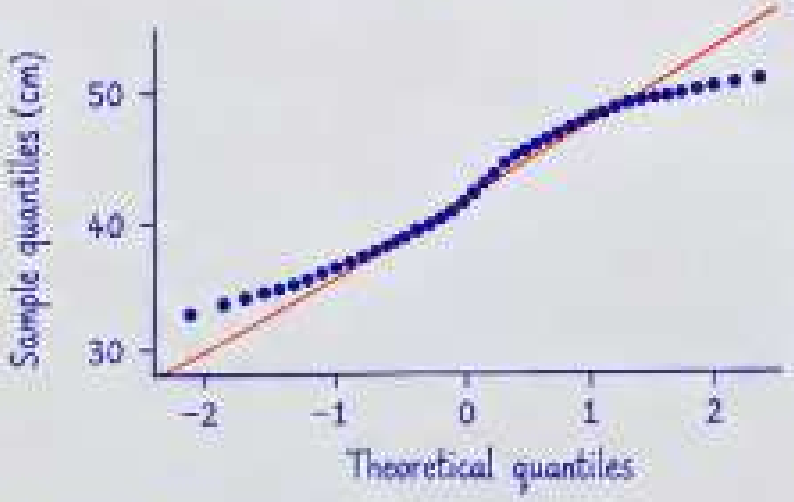
Normal? (Yes / No)	No
Shape (from the list)	Non-normal (possible bimodal)
Evidence in the plot	Points show two clusters and deviate from the line, especially in the middle.

Chart 3 : lab experiment duration (minutes), n = 30



Normal? (Yes / No)	No
Shape (from the list)	Left-skewed
Evidence in the plot	Lower tail is below the line and upper tail is above the line.

Chart 4 : plant height (cm), n = 100



Normal? (Yes / No)	No
Shape (from the list)	Light-tailed
Evidence in the plot	Middle part follows the line but both tails are inside the line (flatter tails).

Part B (4 marks)

B1. Chart 3 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods).

Answer : Departure : Left-skewed.

Remedy : Apply a suitable transformation such as log transformation (or square transformation) to reduce the skewness and make the data closer to normal.

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value :

Answer : `qqnorm(df$value)` + `qqline(df$value)`